

To: The Editor-in-Chief, *Digital Signal Processing* (Elsevier)

Dear Editor,

We are pleased to submit our manuscript, “FreqLite: A Lightweight Frequency-Decomposed Linear Model with Adaptive Reversible Normalization for Robust Long-Term Time-Series Forecasting,” for consideration as an original research article in *Digital Signal Processing*.

Our method is fundamentally a frequency-domain signal-processing approach to time-series forecasting. At its core, a learnable, lossless, partition-of-unity spectral filter (built on the real FFT) decomposes the input signal into frequency bands that are modeled separately and recombined—an adaptive spectral analysis-synthesis pipeline well aligned with the scope of *Digital Signal Processing*. Long-term time-series forecasting underpins decision-making across energy, weather, traffic, and biomedical monitoring, and recent work shows that lightweight linear models can match or exceed far heavier Transformers. Our work advances this direction with a model that is both competitive in accuracy and exceptionally cheap to train and deploy: the entire study runs on a single 4 GB consumer laptop GPU.

The manuscript makes three contributions. First, **FreqLite**, an ultra-lightweight, channel-independent frequency-decomposed linear forecaster that retains and models the high-frequency band rather than discarding it; it is the strongest lightweight model on standard benchmarks and, at long look-back, attains lower average error than a PatchTST Transformer while using about $4\times$ fewer parameters and $2.2\times$ less memory and training time, with improvements that are statistically significant (paired Wilcoxon, $p \approx 10^{-6}$). Second, **A-RevIN**, a regime-adaptive normalization that strictly generalizes RevIN, engaging under non-stationarity (validated on the ILI dataset and a controlled synthetic drift study) and reducing to RevIN without harm otherwise. Third, a **fully reproducible accuracy–efficiency study** on commodity hardware, with all baselines re-implemented and re-run under identical conditions.

We report results honestly, including settings where FreqLite does not lead, and are explicit that the headline strengths on stationary benchmarks are efficiency and the long-lookback result rather than a large accuracy margin. This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the submission; the authors declare no competing interests and received no specific funding.

Thank you for your consideration.

Sincerely,

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